

GROUNDED — the complete system

What it is, everything it does, how it's built, the thinking behind it, and what's next.

Compiled: 2026-05-26 (Claude, Opus 4.7), from the codebase + canonical docs + the Hub/Tracker/Nodes integration work. **Status of this doc:** the definitive overview. Canonical sub-sources where they go deeper: [AGENTS.md](#) (agent/tool spec), [HUB_TRACKER_NODES_PLAN.md](#) (Hub/Tracker/Nodes design + decisions), [ROADMAP.md](#), [README.md](#).

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1. What GROUNDED is

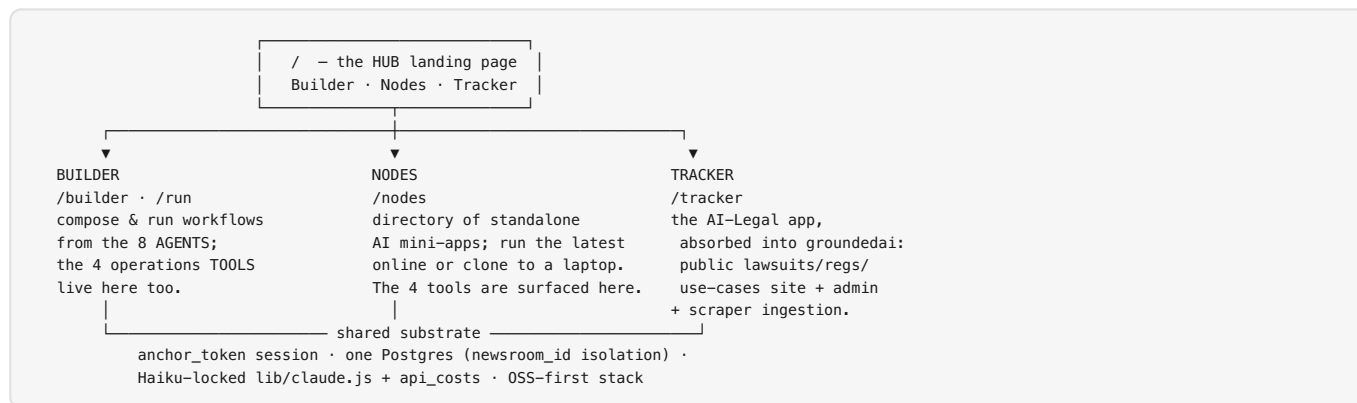
Shared AI infrastructure for African newsrooms. Most African newsrooms have done an *AI workshop*; almost none have working *AI infrastructure*. GROUNDED closes that gap: a multi-tenant web platform where one person at a newsroom (the "AI champion") composes AI workflows from prebuilt agents, and the rest of the team uses what the champion ships through a simple web UI. Sensitive material can run on the newsroom's own hardware; everything else uses Anthropic Claude in the cloud. Apache-2.0, built to outlive its maker — any newsroom or successor team can fork, inspect, and run it.

It is the flagship of **GROUNDED**, a practice under **Develop AI**.

Who	African newsrooms — pilot of 5 ZimZam newsrooms: Capital FM Lusaka, EnviroPress, MakanDay, Maricho Media, VicFallsLive (Zambia + Zimbabwe).
Markets	South Africa, Zimbabwe, Zambia, Kenya (SA + KE in the next cohort).
Scale ambition	5 → 120+ newsrooms across the continent.
Timeline	Build window Jul–Dec 2026; grantee charging from Jan 2027.
Licence	Apache-2.0 — forkable, designed to outlive Develop AI.
Only paid dependency	Anthropic Claude. Everything non-LLM is OSS and fully free at any volume.
Network effect	A cross-newsroom shared workflow library — "what works in Lusaka can run in Harare without anyone rebuilding it." Workflows are framed as <i>solved problems</i> , not pipelines.

2. The shape

A top-level **Hub** with three pillars:



- **Builder** — the existing GROUNDED platform: compose workflows from the **8 journalism agents**; the **4 operations tools** ship alongside. Two modes (champion builds; team uses).
- **Nodes** — an open directory of standalone AI mini-apps newsrooms create and share; each runs **online** in GROUNDED *or* **cloned on a laptop** (one codebase, two runtimes). The 4 tools are surfaced here too (they're newsroom utilities).
- **Tracker** — the "AI-Legal" application (lawsuits / regulations / AI-in-media use-cases across jurisdictions), **absorbed into the GROUNDED repo** and running on the shared database.

3. The thinking & methods

The design philosophy that shaped every decision — these are *constraints*, not preferences.

Product thinking

- **Infrastructure, not a chatbot.** The value is prebuilt, composable, shareable workflows a non-technical newsroom lead assembles — captured as durable assets, not improvised in chat.
- **Workflows are products framed as problems.** Every workflow carries `problem_statement` / `problem_category` / `user_instructions`. User mode groups by *problem*, not by agent. The shared library is a marketplace of solved problems.
- **Two modes.** *Builder mode*: the AI champion composes on a drag-and-drop canvas (React Flow), or describes a workflow in English and Claude drafts the graph ("Describe & build"). *User mode*: the team picks a named workflow and runs it — never sees agents or prompts. **Web only** for journalists.
- **Owned data over scraped data.** Agents/nodes lean on the newsroom's *own* archive, analytics, and history — the signal that's defensible and private.
- **Built to outlive the maker.** Apache-2.0; the method lives in saved workflows inside each newsroom.

Locked technical decisions (do not relitigate without explicit approval)

1. **Haiku 4.5 only** for every Claude call. Hardcoded `GROUNDED_MODEL = 'claude-haiku-4-5-20251001'` in `lib/claude.js`: no model param, no env override, no UI knob. *Why*: cost is the binding constraint at per-newsroom volume; quality suffices (a Researcher benchmark put Haiku at 22x Opus's cost-efficiency on identical extraction). **One exception**: an *availability* fallback to local Ollama (`gemma3:12b`) when Anthropic is unreachable — responses carry `usedFallback:true`; not a cost/tier knob.
2. **OSS + fully-free for every non-LLM module.** No free-tier hosted APIs. Anthropic is the only paid dependency. (Embeddings, translation, STT/TTS, parsing all run in-process or as local binaries.)
3. **Per-newsroom isolation by construction.** Every multi-tenant row has `newsroom_id`; every query is scoped; the `db_read` agentic tool refuses cross-newsroom reads. (AI-Legal *reference* data — lawsuits/regs — is the documented exception: global, `newsroom_id IS NULL`, the same worldwide.)
4. **Anchor** → **GROUNDED** rename is **partial on purpose**. Slugs (`drafter`, `distributor`, ...), table prefixes, the `anchor_token` cookie, the `admin@anchor.local` dev login are intentionally left to protect saved data + live sessions. Don't "fix" them.
5. **WhatsApp is for audiences, not journalists.** Journalists use the web app; WhatsApp is a Digital News Gatherer audience channel (post-pilot).

Build discipline (how the work is done)

- **Step** → **confirm** → **next step**. Each step is one commit the maintainer reviews; nothing chains without a confirm. Migrations are applied by the maintainer (the DB confirm gate).
- **Additive, reversible, investigate-before-writing, no greenfield rewrites.** Look at how prior codebases (Surepath/Holly) did it; log lifts in `REUSE.md`.
- **Verify by running it.** This session caught 4 real bugs only a live run surfaces (a Next `__dirname` path bug, a `BIGINT` → string metrics bug, FK ordering, dep entanglement) — typecheck-clean is necessary, not sufficient.
- **Honest scoping; park side-tasks.** Flag when something's bigger than it looks; defer tangents.

Scope decisions settled during integration (2026-05-25→26)

- **8 agents + 4 tools** (not "11 agents", not "5 tools"). The AI-Legal Tracker is the **Tracker pillar**, not an operations tool.
- **Nodes stay as separate canonical repos**, vendored into GROUNDED as git submodules and cloned/updated independently by newsrooms.
- **Tracker is absorbed into GROUNDED** (its code lives in-repo), not a submodule — and trimmed to the **AI-Legal slice**: the old "Holly" CRM/cohort/training + outreach were dropped; fundraising/ai-assistant/auth/knowledge are covered by GROUNDED's own features.

4. Pillar 1 — Builder

The mature core platform. Authoring at `/builder`; User mode at `/run`. Agents register through `lib/agents/registry.js` with a `category: 'agent' | 'tool'` field; the palette + nav group by category.

The 8 agents (journalism work)

#	Display name	Slug	What it does
1	Verifier	verifier	Multi-source claim verification vs external sources + the newsroom archive; confidence + evidence + citations + gaps; "Never Accuse." Africa-grounded credibility map (SA+ZW+ZM+KE). Agentic enrichment for low-confidence claims.
2	Social media listener	social_listener	Tracks cross-platform posts for damaging narratives; attributes origin (esp. state-aligned actors). OSS lang-detect + NER (Transformers.js).
3	Archivist	archivist	Semantic search + knowledge graph over the newsroom's own archive (BGE-M3 + pgvector). Private per newsroom; powers other agents' archive lookups.
4	Copywriter	drafter	Social copy, headlines, newsletter blurbs, scripts in house style; posts to socials via encrypted per-newsroom credentials.
5	Researcher	researcher	Pulls + scrapes public records, court filings, regulatory + financial docs → research dossiers. Agentic gap-chasing via <code>web_fetch</code> + archive cross-ref.
6	Translator	translator	English ↔ African languages with per-newsroom glossary, multi-model routing (Helsinki opus-mt / NLLB-200 / Masakhane), phrase-level confidence, edit-feedback loop. SA-focused languages.
7	Audio & Video Producer	producer	Radio scripts, podcast outlines, video briefs, audio assembly (Whisper + Piper), audiograms, vertical video, MP4 + editable timeline.
8	Digital News Gatherer	distributor	Inbound only. Triage tips/submissions/contributor pieces (WhatsApp/web/tip lines) into one editor queue; editor routes each to Verifier / Researcher / Operations.

Display-name ≠ slug is deliberate (locked rule #4): names follow `AGENTS.md`; slugs stay as written so saved workflows don't break.

The Builder engine (what's underneath)

- **Workflow registry + runner** — `lib/workflows/{runner,validate,generate,starters}.js`; CRUD + run at `app/api/workflows/*`; per-agent invocations recorded as `workflow_runs` under a `workflow_executions` parent. Per-workflow team assignment (`workflow_assignments`).
- **"Describe & build"** — one Haiku call drafts a valid agent graph from plain English, validated against the live registry (`lib/workflows/generate.js`).
- **Agentic agents (V2)** — Verifier, Researcher, Operations can run a **bounded tool-use loop** (`lib/agents/agentic/loop.js`) with four tools: `archive_search`, `web_fetch`, `invoke_agent`, `db_read` (auto-scoped to the newsroom). Trace viewer at `/observatory/runs/[id]`.
- **Sensitivity classifier + routing (V2)** — `lib/sensitivity/classify.js` deterministically labels input `public | internal | sensitive` (no Claude call); `lib/agents/route.js` decides cloud vs appliance. Escape hatch `GROUNDED_SENSITIVITY_ROUTING=off`.
- **Newsroom appliance (V2)** — signed-HTTPS dispatch so sensitive jobs run on the newsroom's own hardware (Mac mini / NUC / Debian via Ollama). `appliances/`, `lib/appliance/*`, one-script installer. **The execution backend is currently a STUB** (acknowledges dispatches; real Ollama execution is outstanding).
- **Observatory (V2)** — per-workflow + per-edit telemetry; `output_edits` records human Accept/Edit/Reject/Fork. `app/observatory/`.
- **Mentorship dashboard (V2)** — per-team + cohort views over Observatory data, with a k-anonymity floor for the cohort view.
- **Newsroom Profile** — first-class object (strengths, prior coverage, audience data, impact stories, style fingerprint, AI-crawler policy, trusted sources). Read by Fundraiser, Audience, Copywriter.

5. The 4 operations tools

Ship *with* GROUNDED; support newsroom operations rather than journalism. Registered like agents (`category: 'tool'`), each with its own workspace route, draggable into workflows, and **surfaced in the Nodes directory** (they're newsroom utilities used directly).

#	Tool	Slug	What it does
1	Fundraiser	fundraiser	Grant-writing: live funder library + newsroom profile → first-draft applications with budget scaffolding; surfaces cohort joint-application opportunities.
2	Audience Analytics Manager	audience	Analytics ingest (Plausible/Umami/GA/WordPress/CSV) + an AI query layer; headline test + angle sense-check against history.
3	Operations Manager	operations	Whole-org ops: editorial calendar, deadlines, freelancers, sales, logistics, finance, metrics, contributor management. Agentic ad-hoc questions read live tables via <code>db_read</code> .
4	Digital Security Audit	security_audit	A newsroom audits its own setup: external-tool inventory + risk-scoring vs a jurisdiction pack (SA deep; ZW/ZM/KE/TZ/UG/GH/NG/EU/US light) + 90-day routing-history rollout + a Haiku-drafted fix list. Saved + JSON/MD-exportable reports.

Not a 5th tool: the AI Legal/Ethics/Regulation Tracker is the **Tracker pillar** (§7). A thin `legal_tracker` shell + the `/learning` 7-tab UX remain in the Builder registry as a placeholder pending reconciliation with the absorbed Tracker.

6. Pillar 2 — Nodes

An open, growing directory of AI functions for newsrooms. A Node is **mostly standalone** — run the latest **online** inside GROUNDED, or **clone the repo** to run on a laptop. One codebase, two runtimes.

The dual-runtime model (the core idea)

A Node's only application code is its `lib/*.js`, written against a **host interface**. The same code runs:

- **Online (in GROUNDED):** on `lib/nodes/host.js` (`createNodeHost`) — Postgres-backed, session-scoped.
- **Local (laptop):** on `createLiteHost` from the `grounded-node-runtime` package — JSON-file-backed.

A fix in the canonical repo flows to the live site on the next submodule bump, and to every newsroom on their next `git pull`.

The **host facade** mirrors `createLiteHost` exactly so node code runs unchanged:

- `host.ctx` { `newsroomId`, `userId`, `role` } from the session
- `host.tablePrefix` = `node_<slug>`; every query stays inside it
- `host.db.query` (auto-binds `$1 = newsroom_id`) + `host.db.tx`
- `host.ai.chat` (routes through `lib/claude.js`, **Haiku-locked**, logs to `api_costs`)
- `host.parse.docxToHtml` (mammoth)
- `host.log.run/edit/error` + `host.feedback.submit` → the `node_<slug>_{activity,errors,feedback}` tables

The Nodes (`docs/nodes/registry.yaml`)

Slug	Name	Status	Online?	Notes
makanday-analytics	MakanDay Audience Signal	pilot	Yes — live	Owned-data FB engagement-rate analytics; runs at <code>/nodes/makanday-analytics</code> , verified on its real 120-story dataset.
capitalfm-verifier	Capital FM Claim Check	build	standalone	Election-misinformation claim verification. Persists state to local JSON → needs a node-side <code>host.db</code> refactor before it graduates online.
podcasting	Podcast Studio	build	standalone	ElevenLabs voice-cloning podcast studio. JSON/disk state + ElevenLabs SDK → node-side refactor before online.

Vendored as **git submodules** under `nodes/<slug>/` (canonical repos `pauldevelapai/node-*`). Bump with `git submodule update --remote`.

Telemetry (tracking installs + usage)

- **Collector** (Cloudflare Worker, `collector/`): nodes POST `install` / `event` / `feedback` → Airtable (Node Installs / Events / Feedback). Keeps the Airtable token out of public node repos.
- **Harvest** (`harvest/harvest.mjs`): walks each newsroom's fork, pulls committed telemetry, aggregates into the static cohort dashboard (`dashboard/`).
- Online runs also log via `host.log` → the `node_<slug>_activity` tables.

The directory + graduation model

- `/nodes` (`app/nodes/page.tsx`) reads the registry and lists each node with **Run online** (when mounted) + **Clone & run locally** (repo URL + quickstart), plus a "Ships with GROUNDED" section surfacing the 4 tools.
- **Graduation, re-cast (2026-05-25):** a Node is a first-class home in its own right; an agent *optionally* "plugs into" a Node when the job matches (e.g. Verifier ↔ a verification node). Most nodes are unlinked. "Graduation" (node → agent) is the special case, not the rule.

7. Pillar 3 — Tracker (the AI-Legal app, absorbed)

The "AI-Legal" application — a public tracker of AI lawsuits, regulations, and AI-in-media use-cases across jurisdictions, plus an admin back office and an automated ingestion pipeline. Originally a standalone Vite SPA + Express app (the legacy "Holly"/"AI Legal" codebase, ~48 routes / 106 components). **Now absorbed into GROUNDED** and running on the shared database, trimmed to the AI-Legal slice.

What was kept / dropped / merged (decided 2026-05-25)

- **Kept (brought in):** the AI-Legal core + ingestion — `lawsuits`, `regulations`, `usecases`, `legal-sources`, the public site + SSR OG pages, the scraper/triage pipeline, the industry-intelligence feed.
- **Dropped:** the old "Holly" CRM/cohort/training (contacts, organisations, cohorts, courses, learning-*, assessments, ...) and outreach/comms (outreach, social-posts, newsletter, gmail).
- **Merged onto GROUNDED's own:** fundraising → Fundraiser; ai-assistant → agents+runner; `knowledge` → Archivist; auth → the session bridge; dashboard/uploads/jobs/notifications/feedback → GROUNDED equivalents.

How it's wired (the absorption, 5 stages — all done)

1. **Schema** — `db/migrations/039_tracker_import.sql` (generated by `scripts/tracker/build-import.js` from the app's `pg_dump`, trimmed to **16 AI-Legal tables**; FKs to dropped tables stripped; AI-Legal core tables imported **as-is**, unprefixed, global). Data loads one-shot via `--data`.
2. **Service** — the trimmed Express app lives in `tracker/server/` and runs against the shared Postgres (`DATABASE_URL`), started with `npm run tracker` (on `:3055`).
3. **Mount** — `next.config.js` rewrites `/tracker/*` → the service (`TRACKER_URL`, default `:3055`), stripping the prefix; the Hub's Tracker card links to `/tracker/legal`.
4. **SPA front** — the React/Vite SPA in `tracker/client/` builds with `base: '/tracker/'`, served by the service from `client/dist` (static + SPA fallback); `publicFetch / useApi prefix BASE_URL` so calls hit `/tracker/api/*`; router `basename` is `/tracker`; bare `/tracker` → `/legal`.
5. **Auth bridge (H2)** — `tracker/server/middleware/auth.js` verifies GROUNDED's `anchor_token` JWT with the shared `JWT_SECRET` → `req.user`. **One sign-in:** an admin logged into GROUNDED uses the Tracker admin with no second login. Admin routes (`/api/usecases`, `/api/lawsuits`, `/api/regulations`, `/api/legal-sources`) are gated by it.

Ingestion (keeps the tracker live)

`tracker/server/services/legal-ingest/*` (courtlister, RSS/HTML/puppeteer/bluesky/mastodon scrapers, triage, deep-research, insights) + `web-scraper` + `background-jobs` (its gmail/newsletter path is stubbed out — dropped). Scrapers run **on-demand from the admin routes**; the nightly `node-cron` scheduler is intentionally **not auto-started** (avoid surprise scraping/cost).

Verified live

Public API serves the real imported data — **53 lawsuits / 22 regulations / 10 use-cases**; SSR detail pages render ("Disney v. Minimax"); admin routes return 401 without / 200 with the grounded session.

8. The Hub

The shared shell all three pillars hang off (`app/page.tsx`):

- **Landing** — three pillar cards (Builder live; Nodes → `/nodes`; Tracker → `/tracker/legal`), gated by `getCurrentSession()`, role-aware (a user lands in `/run`, builders/admins in `/builder`). Signed-out → `/login`.
- **Auth/session** — GROUNDED's `anchor_token` cookie + `getCurrentSession()` is the single identity; the Tracker's legacy auth is dropped (bridged).
- **Nav** — `app/components/GlobalNav.tsx` groups "Agents — journalism work" and "Tools — newsroom operations".

9. Cross-cutting infrastructure

Data model

- **One Postgres**, raw SQL migrations, numbered `001 – 043`, applied in order by `db/migrate.js` (idempotent, by filename). Needs `pgcrypto` + `pgvector`. Dev DB: port **5433**, database `anchor`.
- **Per-newsroom isolation:** every multi-tenant row has `newsroom_id` `UUID NOT NULL REFERENCES newsrooms(id) ON DELETE CASCADE`.
- **Table families:** `archive_*`, `research_*`, `translation_*`, `audience_*`, `fundraiser_*`, `producer_*`, `distribution_*`, `ops_*`, `social_*`, `verifier_*`, `learning_updates`, `workflow_executions/workflow_runs`, `output_edits`, `api_costs`, `newsroom_appliances/appliance_dispatches/sensitivity_*`, `node_<slug>_*` (migs 040–043), and the imported AI-Legal `ai_*` + `tracker_industry_intelligence` (mig 039).

AI spend

- **Haiku 4.5 only** (`lib/claude.js`); every `chat()` logs a row to `api_costs`. Embeddings are local (\$0). Availability fallback to Ollama `gemma3:12b` (`usedFallback:true` ; disable with `GROUNDED_OLLAMA_FALLBACK_DISABLED=1`).
- The absorbed Tracker has its own `services/claude.js` (still Sonnet-era) — routing its AI calls through `lib/claude.js` + `api_costs` is an open follow-up (T2).

OSS-first non-LLM stack (Anthropic is the only paid dep)

- **Embeddings:** BAAI/bge-m3 via `@huggingface/transformers`, in-process, 1024-dim, pgvector.
- **Translation:** Helsinki-NLP `opus-mt-*` + Meta NLLB-200 distilled.
- **STT/TTS:** Whisper-base ONNX + Piper / `espeak-ng` / macOS `say`.
- **Doc parsing:** `pdf-parse` + `mammoth`.
- **Appliance LLM:** Ollama `gemma3:12b` (sensitive jobs only).
- **Tracker SPA:** React 19 + Vite; node embeddings via `@xenova/transformers`; scraping via `puppeteer` + `cheerio`.

Storage & jobs

- Uploads: local disk under `storage/` in pilot; S3 + Drive mirror are post-pilot.
- Job queue: `pg-boss` on the same Postgres (`scripts/jobs/worker.js`).

10. Stack & conventions

- **Node.js** root, `"type":"commonjs"` (`require / module.exports`). **Next.js 15 App Router** under `app/` (TypeScript; `@/*` → repo root). React 19; canvas via `@xyflow/react`. ESM-only deps via dynamic `import()`.
- **Ports:** GROUNDED 3002; absorbed Tracker service 3055.
- **Auth:** roll-our-own JWT + `bcryptjs`, `httpOnly` cookie `anchor_token` (`jwt.sign({sub,nrm,role}, JWT_SECRET, {7d})`).
- **Tests:** `node --test`, single-concurrency (`npm test`); the node host facade has 11 hermetic unit tests.
- **TS config:** `allowJs`, `moduleResolution: bundler`; reads resolve from `process.cwd()` at runtime (NOT `__dirname` — that points into `.next/server/*` in the bundle).
- **Dev gotchas (maintainer's Mac):** Postgres on :5433 (peer auth); `unset ANTHROPIC_API_KEY` before `npm run dev` (the shell exports an empty string that shadows `.env`); `unset NODE_ENV` before `npm install`. (One shell has a `zsh` global alias that expands `ANTHROPIC_API_KEY` to its value — don't put the bare name on a command line there.)

11. Repo layout

<code>app/</code>	Next.js 15 (Hub, Builder /builder, User /run, /nodes, agent/tool workspaces, observatory, mentorship, appliances; <code>app/nodes/makanday-analytics</code> = the live node mount)
<code>lib/</code>	<code>claude.js</code> (Haiku lock), <code>db.js</code> , <code>auth.js</code> , <code>agents/</code> (registry + 13 modules + <code>agentic()</code>), <code>nodes/host.js</code> (the host facade) + <code>nodes/registry.js</code> , <code>sensitivity/</code> , <code>observatory/</code> , <code>appliance/</code> , + per-agent backends (archive, research, translation, audio, video, ...)
<code>db/migrations/</code>	001-043 SQL (039 = trimmed AI-Legal import; 040-042 node tables; 043 node int metrics)
<code>nodes/<slug>/</code>	the 3 Node submodules (canonical repos)
<code>tracker/</code>	the absorbed AI-Legal app – <code>server/</code> (Express, AI-Legal slice) + <code>client/</code> (React/Vite SPA)
<code>collector/</code>	Cloudflare Worker – node telemetry → <code>Airtable</code>
<code>harvest/</code> + <code>dashboard/</code>	cohort telemetry harvest + static dashboard
<code>appliances/</code>	per-newsroom appliance protocol + installer (execution backend is a STUB)
<code>scripts/tracker/</code>	<code>build-import.js</code> (<code>pg_dump</code> → trimmed migration 039)
<code>docs/</code>	<code>AGENTS.md</code> , <code>ROADMAP.md</code> , <code>HUB_TRACKER_NODES_PLAN.md</code> , this file, <code>nodes/registry.yaml</code> , <code>tracker/overlap-map.yaml</code> , <code>V2_PLAN.md</code> , <code>SECURITY_AUDIT_PLAN.md</code> , ...

12. How to run the whole platform

```
# 1. DB (one-time; already at migration 043 on the dev box)
unset ANTHROPIC_API_KEY && npm run migrate

# 2. Build the Tracker SPA once (dist/ is gitignored)
cd tracker/client && npm install && npm run build && cd ../..

# 3. Run – three processes
unset ANTHROPIC_API_KEY && npm run dev      # GROUNDED (Next) on :3002 – restart to apply the /tracker proxy
npm run tracker                            # absorbed AI-Legal service on :3055 (reads tracker/.env -> shared DB)
unset ANTHROPIC_API_KEY && npm run worker   # pg-boss worker (optional, for async jobs)
```

Sign in (admin@anchor.local / changeme123 on the dev seed) → Hub:

- **Builder** → /builder (champion) or /run (user).
- **Nodes** → /nodes → MakanDay **Run online** (real data).
- **Tracker** → /tracker/legal (public AI-Legal site); admin works with the same session, no second login.

13. Current state

Both pillars moved from "compiles" to "running against the real database" this integration arc.

- **Builder (V1 + V2): shipped.** All 8 agents + 4 tools, drag-and-drop + Describe & build, User mode, Observatory, Mentorship, agentic loop, sensitivity routing, appliance protocol. Two known stubs: appliance *execution*, and the legal_tracker shell.
- **Nodes pillar: live (N1–N5).** Host facade + 11 green tests; submodules vendored; per-node migrations; **MakanDay runs online on its real 120-story dataset**; /nodes directory + Hub wired. (Two bugs caught + fixed by running it: the registry __dirname path; BIGINT→string metrics.)
- **Tracker pillar: absorbed (T0 + T1 x5), verified.** Scope decided, 039 trimmed to 16 tables, the AI-Legal app brought into tracker/, mounted at /tracker, SPA served, auth-bridged to the GROUNDED session, scrapers in. Serving the real 53 lawsuits / 22 regs / 10 use-cases.
- **DB:** migrations through 043 applied; AI-Legal data loaded.
- **Branch:** hub-tracker-nodes (~18 commits), not yet merged to main.

14. What needs to be done next

Immediate / housekeeping

1. **Rotate the Anthropic API key** that was pasted into a terminal earlier (compromised) — and put the new one in .env. Required for any AI feature (node briefs, Tracker AI).
2. **Merge hub-tracker-nodes → main** once reviewed (PR), and confirm a clean run from a fresh checkout.
3. Fix the --data emitter to order parent tables before children (a few AI-Legal child/event tables failed to load on FK ordering).

Tracker follow-ups 4. **T2 — route the Tracker's Claude calls through lib/claude.js + api_costs** (currently its own services/claude.js); decide Tracker model policy (keep its model vs the Haiku lock). 5. Trim the SPA's inert admin/CRM pages (their backends were dropped); decide whether/where the admin nav surfaces. 6. Decide if/when to **auto-start the nightly scraper scheduler** (currently on-demand only). 7. **T3 — reconcile the /learning shell + legal_tracker tool vs the absorbed Tracker** (retire the shell, repoint it, or keep both) — driven by docs/tracker/overlap-map.yaml.

Nodes follow-ups 8. **capitalfm-verifier + podcasting node-side host.db refactors** (in their own repos) so their state graduates to Postgres and they can run online (N4 for those two).

Platform follow-ups (pre-existing) 9. **Real Ollama-backed appliance execution** — the protocol is shipped; the backend is a STUB (appliances/agent-runner/server.js). 10. **Deep-research the light jurisdiction packs** (ZW/ZM/KE/TZ/UG/GH/NG/EU/US) as cohort countries onboard.

Deferred post-pilot (don't pull forward): WhatsApp audience channel; real S3 + Drive mirror; Lightsail deploy; per-newsroom Anthropic keys; cross-cohort search; multi-cohort UI.

15. Source-of-truth map

For...	Read...
Agent + tool product spec	docs/AGENTS.md
Hub/Tracker/Nodes design + the scope decisions	docs/HUB_TRACKER_NODES_PLAN.md
Forward view / outstanding work	docs/ROADMAP.md
Platform overview + locked rules + repo layout	README.md
V2 build history	docs/V2_PLAN.md
Security Audit tool	docs/SECURITY_AUDIT_PLAN.md
Appliance real-execution scope	docs/APPLIANCE_EXECUTION_PLAN.md
Nodes registry	docs/nodes/registry.yaml
Tracker import + overlap catalogue	docs/tracker/overlap-map.yaml
Reuse/lift inventory	REUSE.md